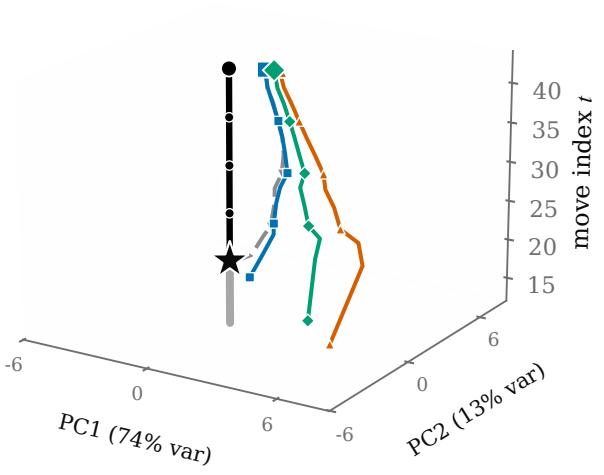
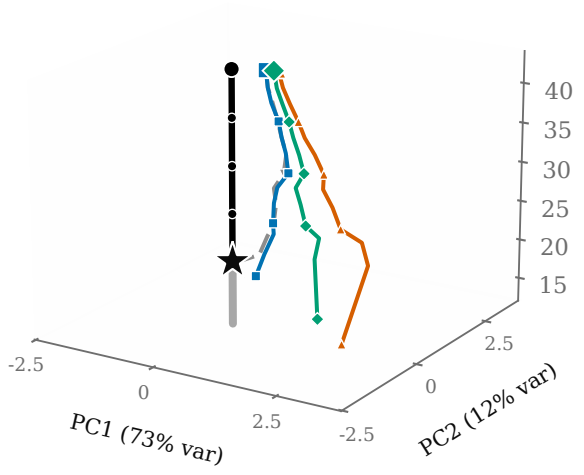
(a) where the branches go: the fork in each representation space, one path per rule
mean over n = 100 games, all forking at t_0 = 20

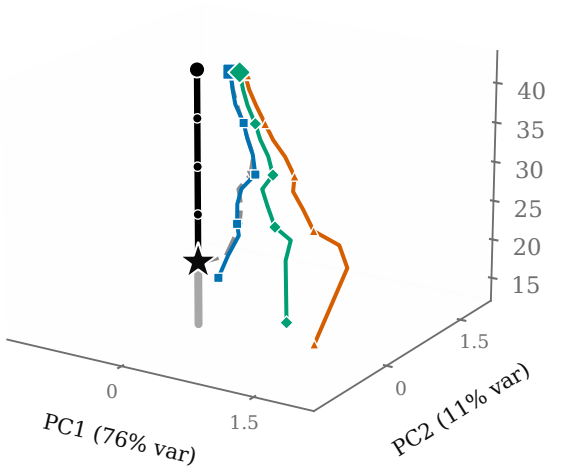
raw layer 5 (reference space)
coherence at t = 44: R1 0.18 · R2 0.22 · R3 0.19



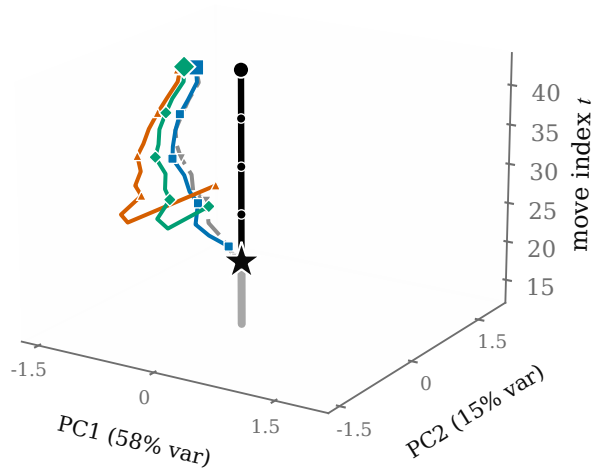
PAnneal (p-anneal, pen 0.042)
coherence at t = 44: R1 0.17 · R2 0.19 · R3 0.17



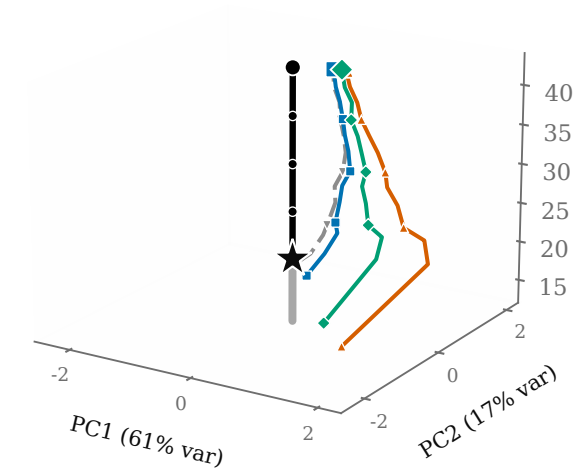
Standard (L1 0.059)
coherence at t = 44: R1 0.16 · R2 0.20 · R3 0.17



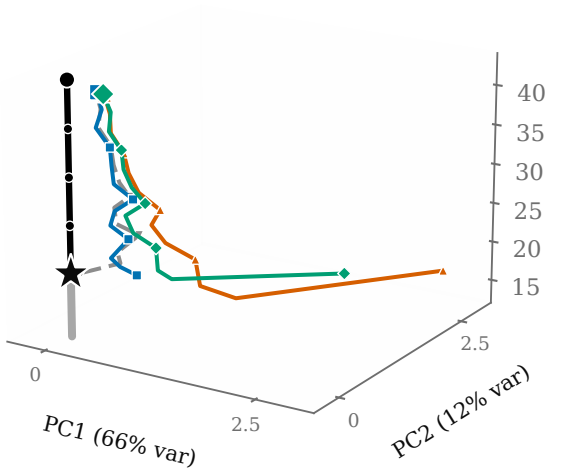
GatedSAE (L1 1.37)
coherence at t = 44: R1 0.14 · R2 0.17 · R3 0.16



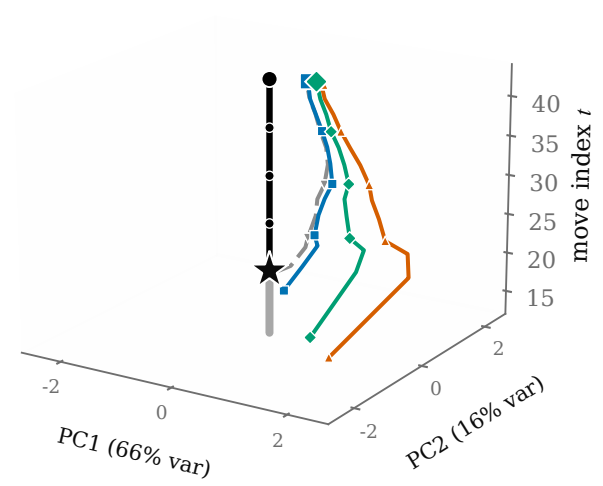
TopK (k=80)
coherence at t = 44: R1 0.15 · R2 0.17 · R3 0.15



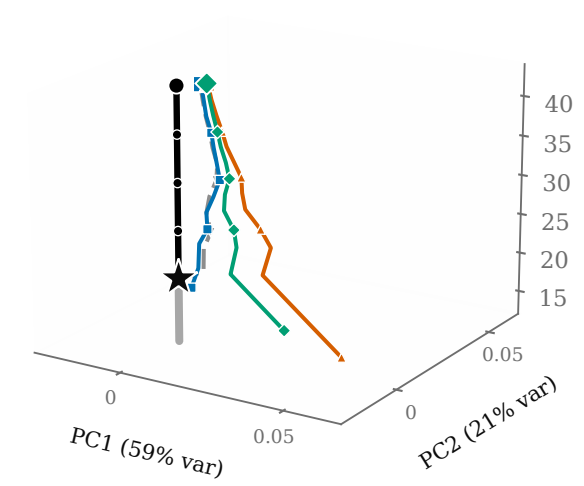
GatedAnneal (p-anneal, pen 0.955)
coherence at t = 44: R1 0.14 · R2 0.15 · R3 0.14



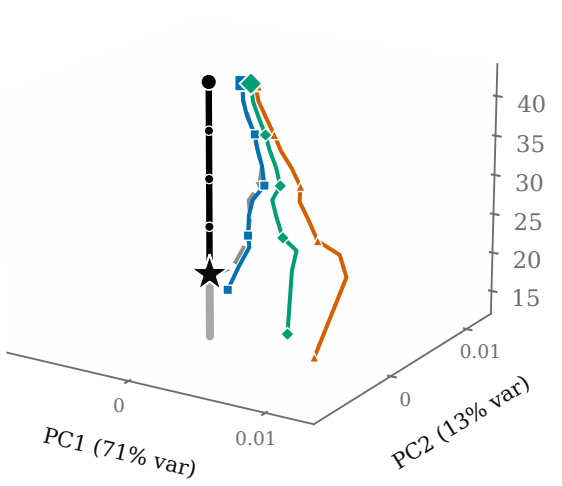
BatchTopK (k=80)
coherence at t = 44: R1 0.16 · R2 0.18 · R3 0.16



CosineSpaDE (γ=1.0)
coherence at t = 44: R1 0.16 · R2 0.17 · R3 0.16



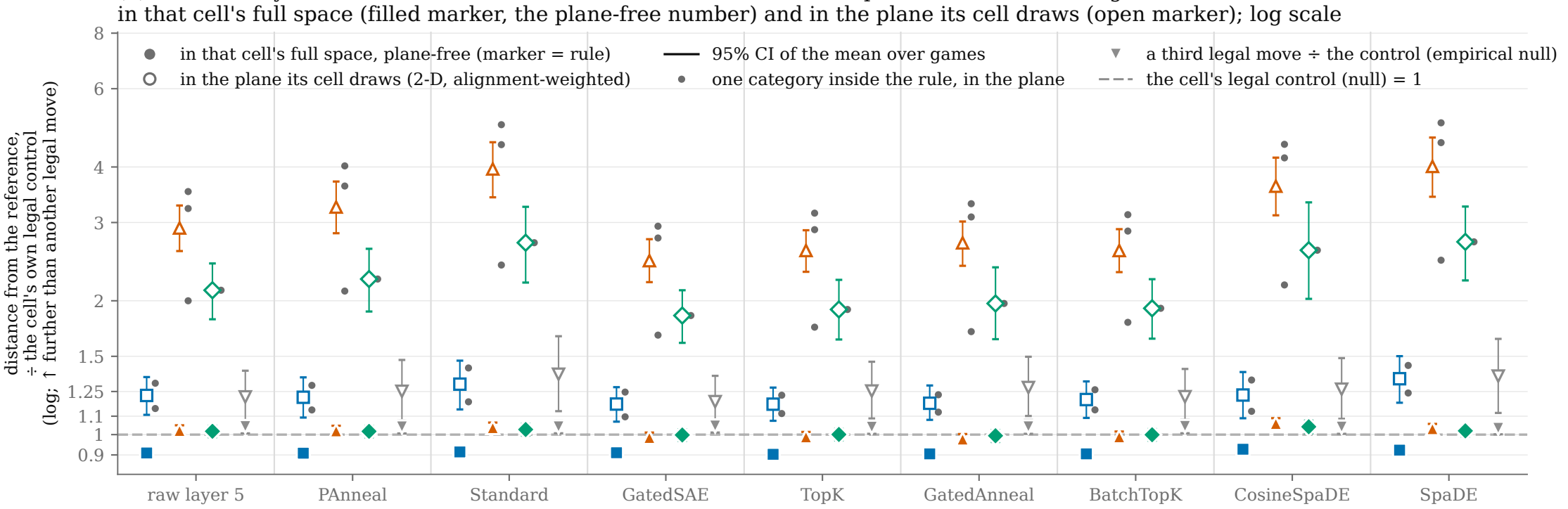
SpaDE (λ=1e-3, γ=1e-4)
coherence at t = 44: R1 0.16 · R2 0.19 · R3 0.17



shared prefix (branches identical) · reference (the game as played) · R1 placement (mean of 2) · R2 bracketing (mean of 3) · R3 turn (mean of 1) · branch point t_0 · control: another legal move (null)

within ÷ between-rule spread in raw layer 5, per game and not the mean field's: median 0.92 [IQR 0.84–0.98] — near 1, the rule grouping is a label more than a finding
coherence, per rule = ‖mean over games of D‖ ÷ mean over games of ‖D‖ for the path (a) draws: the share of one game's displacement that survives (0 = all of it cancels)

(b) how far they run: each rule's distance from the reference as a multiple of the same cell's legal control —
in that cell's full space (filled marker, the plane-free number) and in the plane its cell draws (open marker); log scale



per game: the mean over the 13 drawn offsets r = 0, 2, ..., 24, ÷ that game's own legal control; markers = the mean of those n = 100 per-game ratios (median in the CSV), bars = 95% percentile bootstrap CI over games (2000 resamples, seed 0).
The open marker's plane is fit on these same 100 games' mean field — the direction the illegal branches share — so a distance measured in it is alignment-weighted and not plane-free: that is the whole 1.1–3.9× gap between the two markers

the first 100 of the 1696 branch-complete games forking at t_0 = 20, ascending by game id: no game-level criterion enters this figure, and game 1235 — the argmax the single-game figure draws — is in it as one ordinary member of 100 (README)

(a) averages displacement vectors, so directions that differ between games cancel and each cell prints the surviving share; (b) averages per-game distances, which does not cancel

R1 placement (2): square holds the mover's disc · square holds the opponent's disc

R2 bracketing (3): empty, touches no discs · empty, touches only the mover's discs · empty, touches opponent discs, no line closes

R3 turn (1): flips only for the opponent (turn confusion)